Auto Ticket Classification using Flow Designer

Project Report

Team ID	**SWTID-2026-7310**
Team Size	5
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Team Member	Pradeep B
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1. Introduction

1.1 Project Overview

The school IT helpdesk receives many incident requests every day from students and teachers: Wi-Fi issues, projector failures, password problems and slow computers. IT staff currently review each request and assign a category, which is time-consuming and inefficient.

This project automates ticket classification with ServiceNow Flow Designer. When a ticket is created, the flow analyses keywords in the Short description, sets Category and Subcategory automatically and sends a confirmation email to the caller. Tickets are stored in a custom table with dependent choice fields, and the whole solution is no-code and easy to maintain.

1.2 Purpose

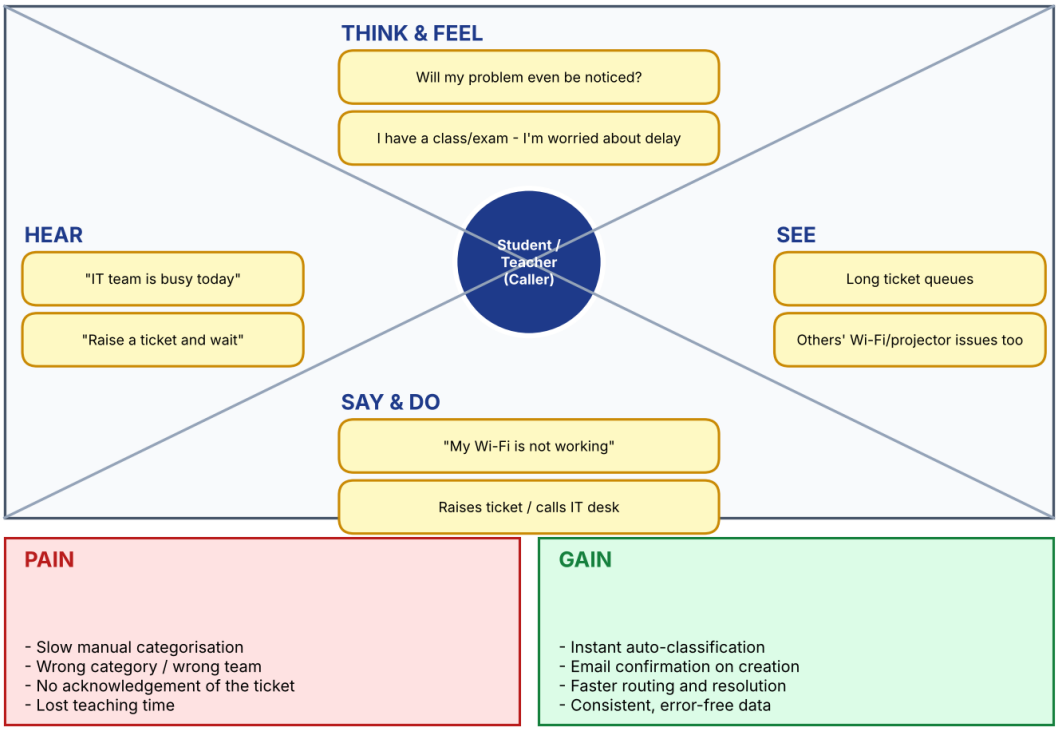
- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

2. Ideation Phase

2.1 Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A student / teacher raising an IT ticket	get my Wi-Fi, projector, password or slow-PC issue fixed quickly	tickets wait for manual categorisation or reach the wrong team	there is no automatic classification or acknowledgement	frustrated and worried about lost class time
PS-2	An IT helpdesk staff member	classify every incident with the right category and subcategory	it is slow and error-prone as requests grow	every ticket is read and classified by hand	overloaded and unable to focus on resolving issues

2.2 Empathy Map Canvas

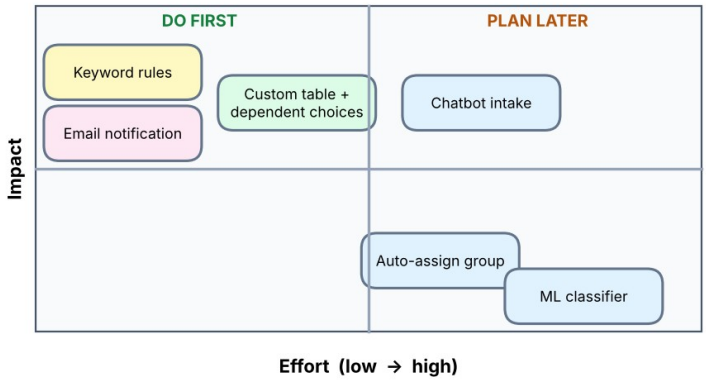


2.3 Brainstorming

Team: Mohan Kumar S (Leader), Muhammed Ibrahim A, Ponakilan J, Pradeep B, Ragulprathap R

Problem statement: How might we classify school IT tickets automatically at the time of creation, so IT staff spend no time on manual categorisation?

Keyword rules	Data model	Notification	Other ideas
Flow Designer If / Else-If on Short description	Custom table Incident WorkFlow	Email caller on ticket creation	ML text classifier (needs training data)
Keywords: Wi-Fi, Projector, Password, Slow	Dependent Category → Subcategory choices	Include ticket number in email	Chatbot intake (Virtual Agent)
			Auto-assign group by category



Selected ideas: keyword-based classification with Flow Designer, custom table with dependent choices and an automatic email to the caller. Automatic group assignment, an ML classifier and chatbot intake are kept for future scope.

3. Requirement Analysis

3.1 Customer Journey Map

	Notice issue	Create ticket	Auto-classify	Receive email	Resolution
Steps	Wi-Fi/projector/ password problem	Enters Caller + Short description	Flow Designer sets Category & Subcategory	Caller gets 'request submitted' mail	IT staff work the correctly routed ticket
Interactions	Teacher / student classroom, lab	Incident WorkFlow form	ServiceNow Flow Designer	Caller's mailbox	IT team / ServiceNow
Goals	Get back to work	Report quickly without forms	Right team gets it	Know it was received	Issue fixed fast
Positive moments	-	Only 2 fields needed	Instant, no waiting	Reassurance	Fast turnaround
Negative (before)	Lost class time	Must pick correct category	Manual review delay	No acknowledgement	Misrouted tickets
Opportunities	Self-service tips	Keyword hints	Auto-assign group, SLA	SMS / Teams alert	Satisfaction survey

3.2 Solution Requirement

Functional requirements

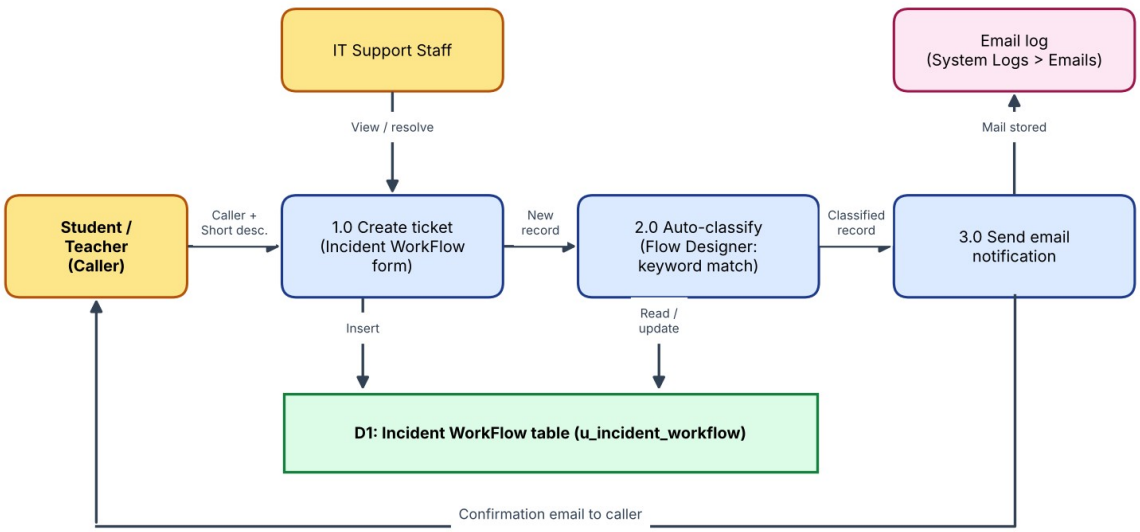
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Ticket Creation	Caller creates a ticket entering Caller and Short description; ticket number generated automatically (INC prefix)
FR-2	Automatic Classification	Flow triggers on record creation when Category is empty; keyword match sets

		Category and Subcategory (Network/Wi-Fi, Hardware/Projector, Access/Forgot Password, Performance/Slow Computer)
FR-3	Category - Subcategory Dependency	Subcategory options change with the selected Category
FR-4	Email Notification	Send Email action mails the caller after classification
FR-5	Structured Data Storage	Custom table Incident WorkFlow (u_incident_workflow) with standard fields
FR-6	Maintenance & Deployment	No-code flow; changes captured in Project Update Set and exported to XML

Non-functional requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Caller enters only Caller and Short description; standard ServiceNow form
NFR-2	Security	Access controls on the custom table; only authorised users modify tickets
NFR-3	Reliability	Rule-based and deterministic: same keyword gives the same category
NFR-4	Performance	Flow runs at record creation and completes within seconds
NFR-5	Availability	Runs on the ServiceNow cloud platform
NFR-6	Scalability	New keywords or categories added by configuration, no coding

3.3 Data Flow Diagram



User stories

User Type	Epic	USN	User Story / Task	Priority	Release

Student / Teacher	Ticket Creation	USN-1	Create an IT ticket by entering name and short description	High	Sprint-3
Student / Teacher	Automatic Classification	USN-2	Ticket categorised automatically	High	Sprint-2
Student / Teacher	Email Notification	USN-3	Receive an email confirming the request was submitted	High	Sprint-2
IT Support Agent	Ticket Handling	USN-4	See classified tickets without manual triage	High	Sprint-2
IT Support Agent	Ticket Handling	USN-5	Set State, Assigned group and Assigned to	Medium	Sprint-1
Administrator	Data Model	USN-6	Create custom table with auto-numbering	High	Sprint-1
Administrator	Data Model	USN-7	Define choice fields and dependent Category > Subcategory	High	Sprint-1
Administrator	Flow Automation	USN-8	Maintain classification rules in Flow Designer without code	Medium	Sprint-2
Administrator	Deployment	USN-9	Capture changes in an update set and export XML	Medium	Sprint-3
Administrator	Security	USN-10	Restrict ticket data to authorised users	Medium	Sprint-3

3.4 Technology Stack

S.No	Component	Description	Technology
1	User Interface	Incident WorkFlow form for callers and agents	ServiceNow UI / Form Designer
2	Application Logic	Trigger, If / Else-If classification, Update Record, Send Email	ServiceNow Flow Designer (no-code)
3	Database	Custom table u_incident_workflow with auto-number, choice and reference fields	ServiceNow platform database
4	File Storage	Update set exported as XML	Local file system / Mail
5	External API / ML model	Not used - rule-based keyword matching	-
6	Infrastructure	Cloud deployment in Global application scope	ServiceNow cloud instance
7	Security	Access control on the custom table (role u_incident_workflow_user)	ServiceNow ACLs / roles

4. Project Design

4.1 Problem Solution Fit

Problem-Solution Fit Canvas - Auto Ticket Classification using Flow Designer

1. CUSTOMER SEGMENT(S)

School students and teachers who raise IT tickets; IT helpdesk staff who triage them.

6. CUSTOMER CONSTRAINTS

No budget for coding / ML tools; non-technical users; limited IT staff; must run inside existing ServiceNow instance.

5. AVAILABLE SOLUTIONS

Manual review and categorisation by IT staff; phone calls and emails to the helpdesk; spreadsheets.

2. JOBS-TO-BE-DONE / PROBLEMS

Get Wi-Fi, projector, password, slow-PC issues fixed quickly; staff need tickets correctly categorised without effort.

9. PROBLEM ROOT CAUSE

Category and subcategory are chosen manually, so ticketing is slow, inconsistent and error-prone as volume grows.

7. BEHAVIOUR

Users describe the problem in a short sentence; they skip or mis-pick categories; IT staff re-open tickets to fix them.

3. TRIGGERS

Class or exam disrupted; repeated misrouted tickets; growing daily ticket count.

10. YOUR SOLUTION

Flow Designer flow triggered on record creation; keyword match on Short description sets Category + Subcategory; email confirms to caller.

8. CHANNELS OF BEHAVIOUR

Online: ServiceNow form, email. Offline: phone call / walking up to IT desk.

4. EMOTIONS: BEFORE / AFTER

Before: frustrated, anxious, ignored.
After: reassured, confident, supported.

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement	Manual, slow and error-prone categorisation of daily school IT tickets
2	Idea / Solution description	Flow Designer flow classifies tickets by Short description keywords, sets Category and Subcategory and emails the caller
3	Novelty / Uniqueness	Fully no-code, rule-based, tailored to common school IT issues, with dependent choices and instant acknowledgement
4	Social Impact / Customer Satisfaction	Faster support, less lost class time, instant reassurance, staff time freed for resolution
5	Business Model (Revenue Model)	Internal service-improvement tool; value from saved staff hours; reusable as an update set
6	Scalability of the Solution	New keywords and categories added by configuration; update set installs on other instances

4.3 Solution Architecture

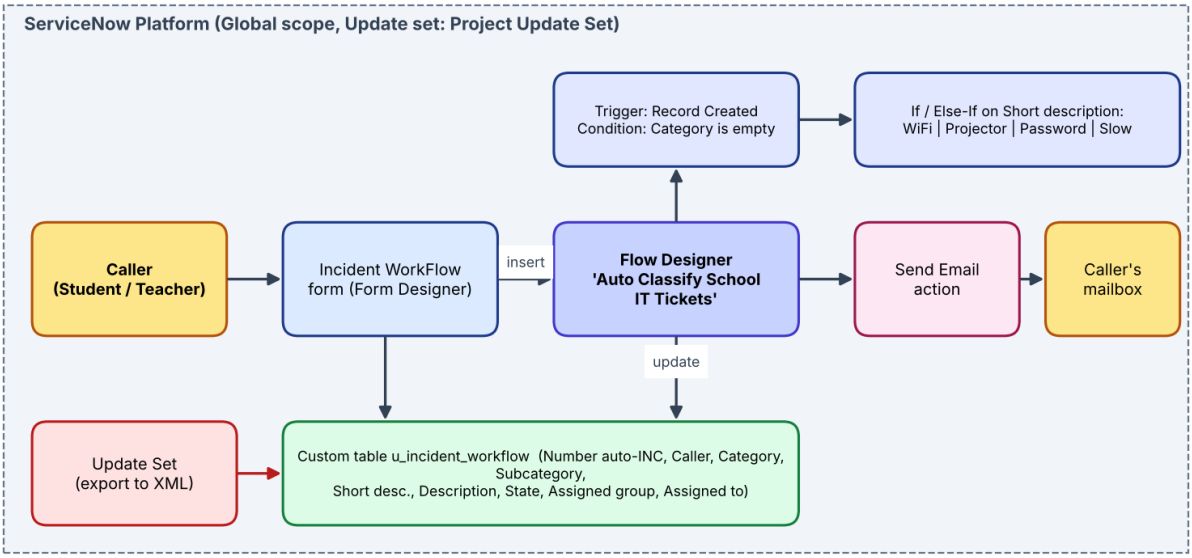


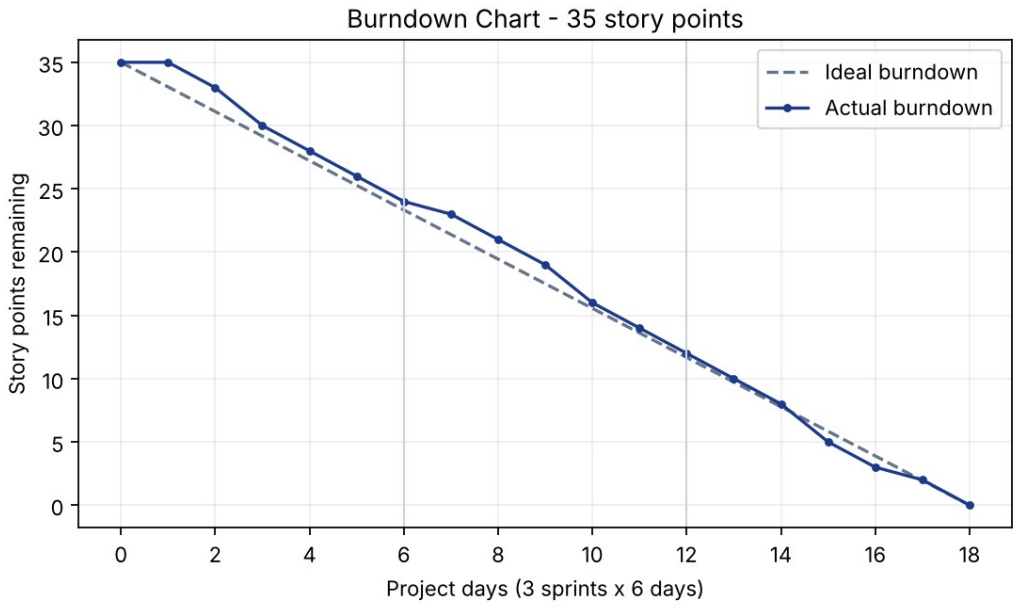
Figure: Architecture and data flow of the Auto Ticket Classification solution

5. Project Planning & Scheduling

5.1 Project Planning

Sprint	Epics / Work	Story Points	Duration	Start	End (Planned)	Completed	Release (Actual)
Sprint-1	Environment setup, Data model (table, fields, choices, dependency)	11	6 Days	19 Jan 2026	24 Jan 2026	11	24 Jan 2026
Sprint-2	Flow automation (trigger, 4 classification branches, email, activate)	12	6 Days	26 Jan 2026	31 Jan 2026	12	31 Jan 2026
Sprint-3	Testing & validation, deployment (update set), documentation	12	6 Days	02 Feb 2026	07 Feb 2026	12	07 Feb 2026

Total story points = 35; velocity = 35 / 3 = 11.67 (about 12 story points per sprint); average velocity per day = 11.67 / 6 = about 1.94.



Burndown chart

6. Functional and Performance Testing

6.1 Performance Testing

The Flow Designer automation was tested end to end. Two classification scenarios (Network / Wi-Fi and Projector) were executed with 100% correct classification; the confirmation email was verified in System Logs > Emails; auto-numbers were generated without duplication (INC00511, INC00512).

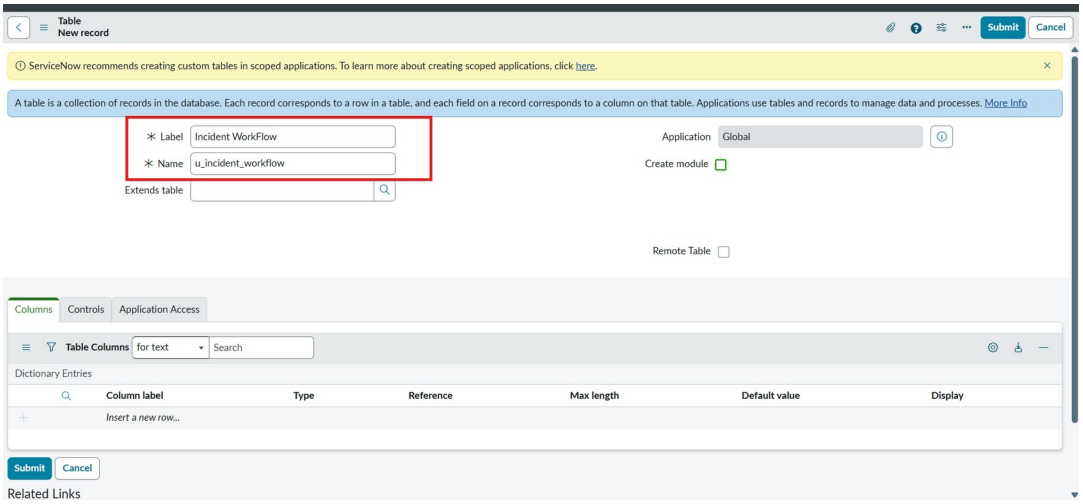
ID	Scenario	Expected	Result
TC-001	Auto-number generated for new tickets	Unique INC number generated (INC prefix)	Pass
TC-002	Subcategory depends on Category	Network > Wi-Fi; Hardware > Projector; Access > Forgot Password; Performance > Slow Computer	Pass
TC-003	Network / Wi-Fi ticket (Test Scenario 1)	Category = Network, Subcategory = Wi-Fi	Pass
TC-004	Projector ticket (Test Scenario 2)	Category = Hardware, Subcategory = Projector	Pass
TC-005	Password ticket	Category = Access, Subcategory = Forgot Password	Not tested
TC-006	Slow computer ticket	Category = Performance, Subcategory = Slow Computer	Not tested
TC-007	Confirmation email to caller	Email 'Your Request for the issue has been submitted.' sent to the caller's email	Pass
TC-008	Only authorised users access ticket data	Access denied	Not tested
TC-009	Update set completed and exported	XML file downloaded	Pass

Open observations from review:

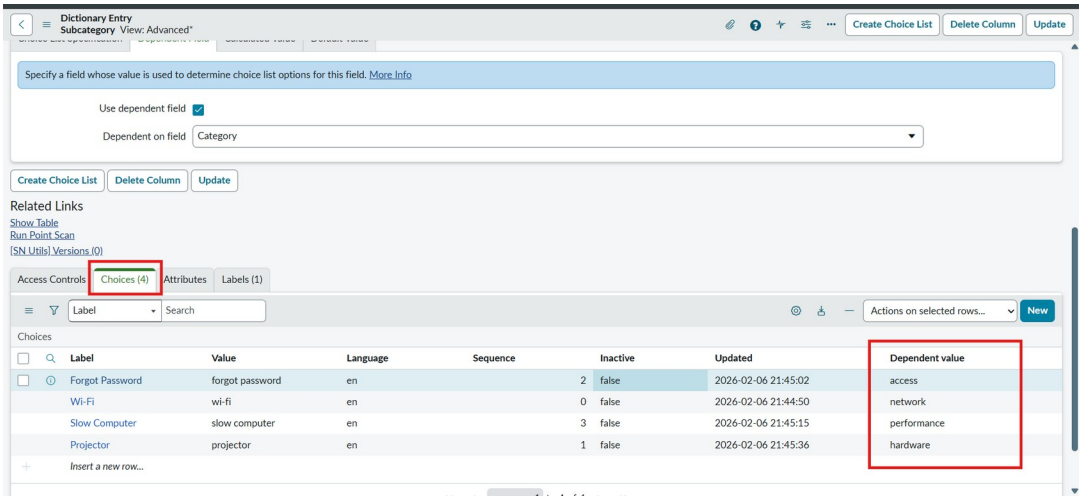
Bug ID	Description	Severity	Status
BG-001	Keyword 'WiFi' (no hyphen) is not matched: the flow condition checks 'Wi-Fi' or 'Network'	Medium	Open
BG-002	Requirement lists 'Login' and 'Hanging' as keywords, but the flow only checks 'Forgot password' and 'Slow computer'	Low	Open

7. Results

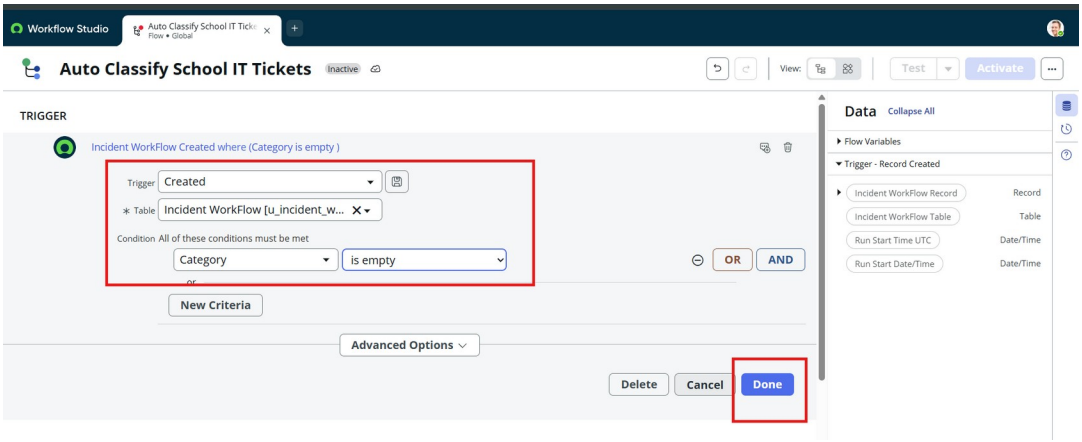
7.1 Output Screenshots



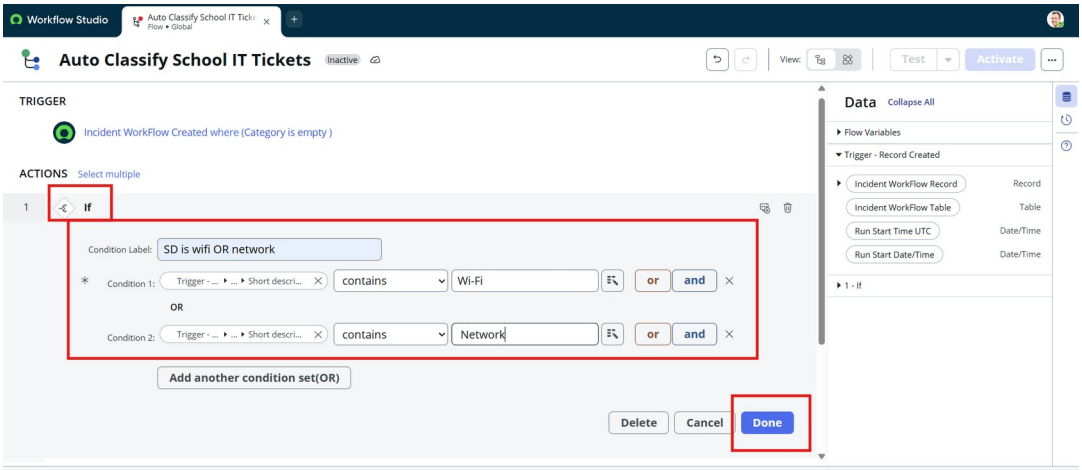
Custom table Incident WorkFlow (u_incident_workflow)



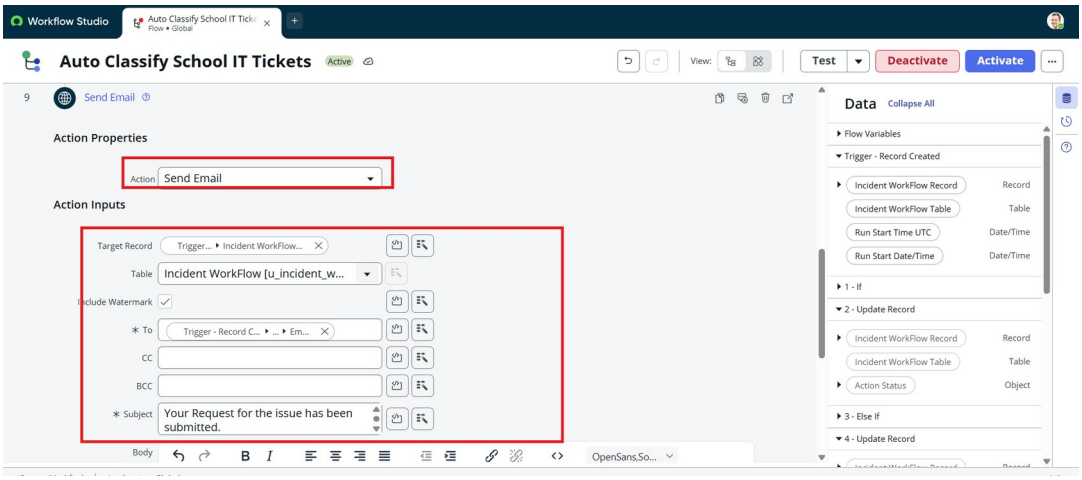
Dependent Category - Subcategory choices



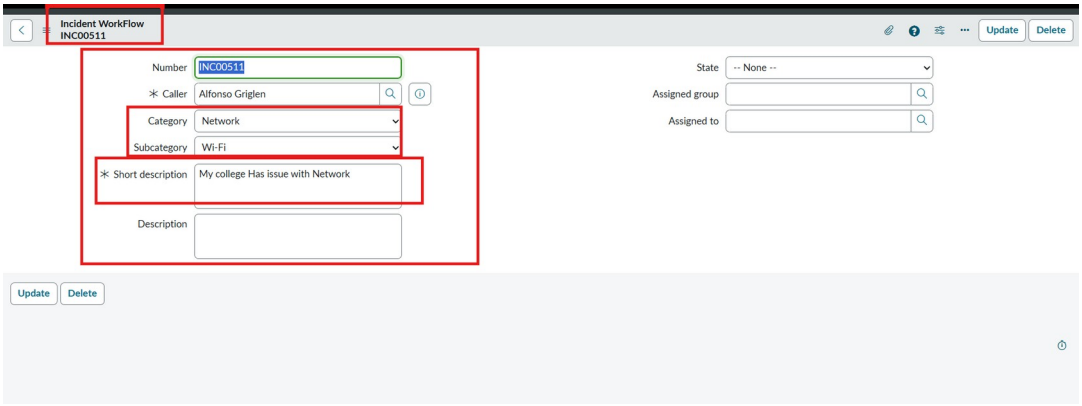
Flow trigger: Record Created, Category is empty



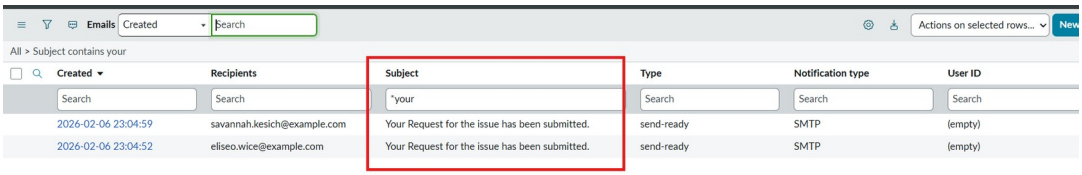
Wi-Fi / Network classification branch



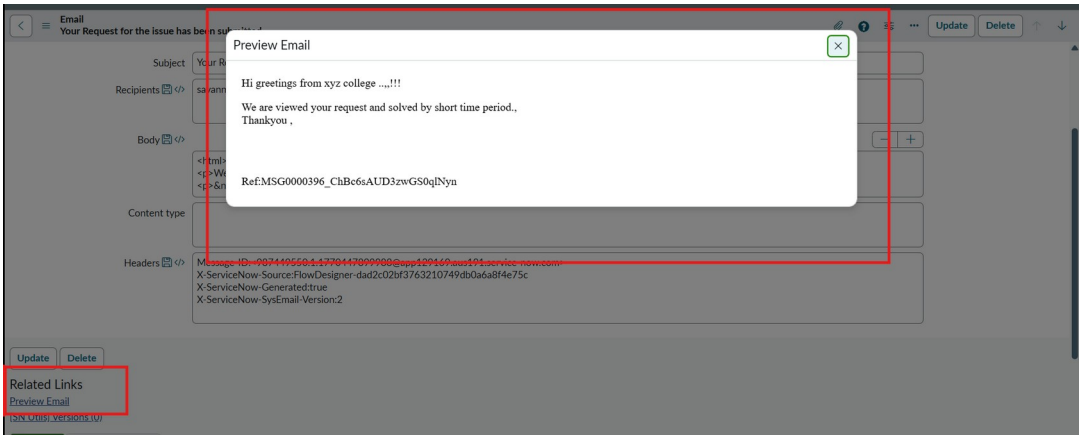
Send Email action to the caller



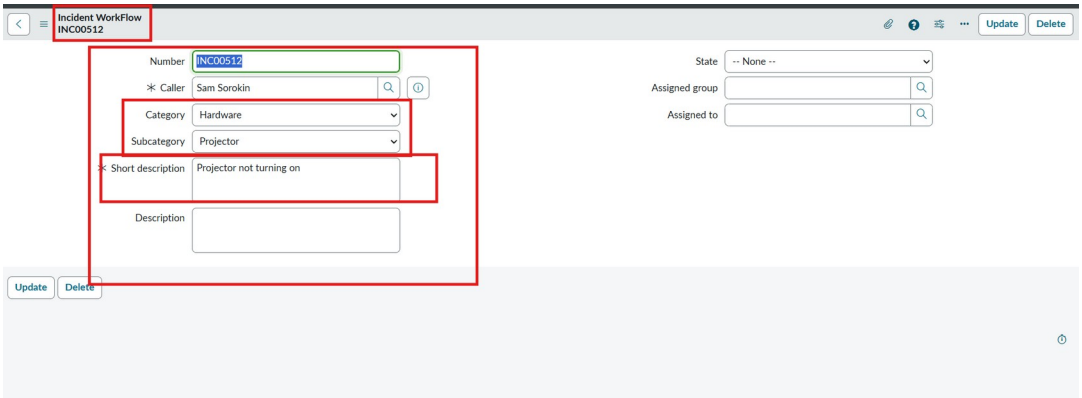
Test Scenario 1 - ticket auto-classified as Network / Wi-Fi



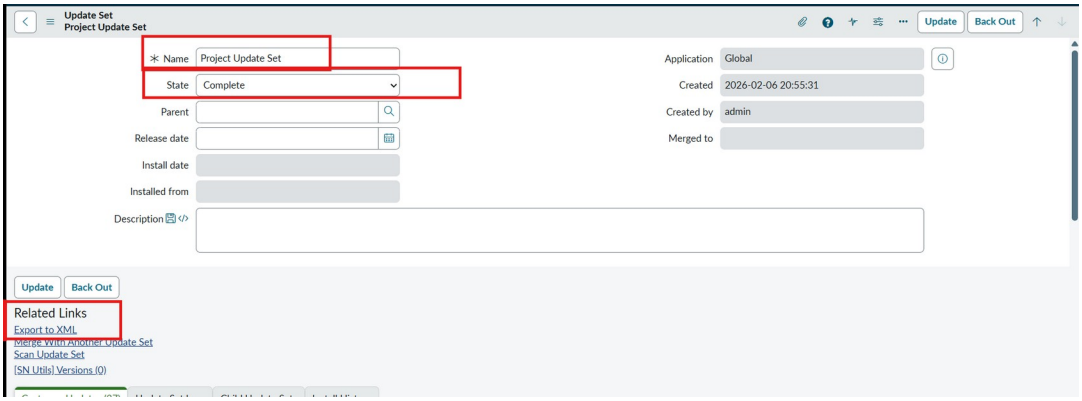
Confirmation emails in System Logs > Emails



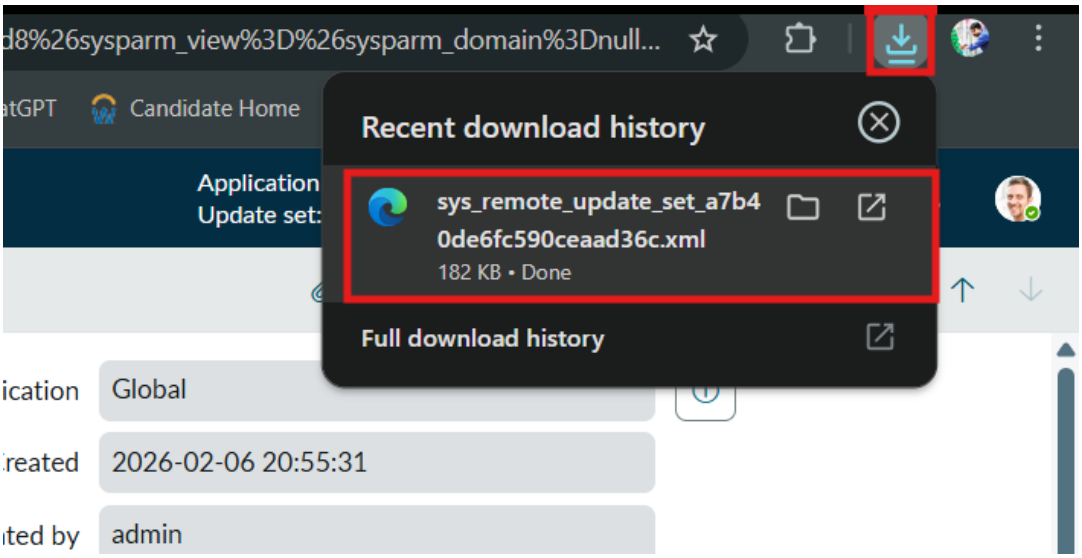
Email preview



Test Scenario 2 - ticket auto-classified as Hardware / Projector



Project Update Set set to Complete



Update set exported as XML

8. Advantages & Disadvantages

Advantages

- Automatic classification at ticket creation removes manual triage.
- No-code: configured entirely in Flow Designer, easy to maintain.
- Consistent categories and standard data format.
- Caller gets an instant confirmation email.
- Easy to extend with new keywords or categories; portable through an update set.

Disadvantages

- Keyword matching is exact: variations such as 'WiFi' vs 'Wi-Fi' or unlisted words are not classified.
- Only four issue types are covered; other issues stay unclassified.
- No automatic assignment to groups or SLA handling yet.

9. Conclusion

The Auto Ticket Classification using Flow Designer project delivers an end-to-end automation solution for a school IT helpdesk. Through structured requirement analysis, careful backend configuration and no-code automation using Flow Designer, the system automatically classifies IT tickets from the short description provided by students. Dependent choice fields ensure data accuracy, and automated email notifications improve communication by acknowledging ticket creation to the caller.

The solution was developed in phases, with testing and security validation confirming reliable operation and data integrity. The design suits small to medium environments such as educational institutions and can be extended with assignment automation, SLA tracking or predictive intelligence.

10. Future Scope

- Automatically assign the Assignment group (and agent) based on category.
- Cover more keywords and variations (for example 'WiFi', 'login', 'hanging') and more issue types.
- SLA tracking, priority and escalation rules.
- Chatbot / Virtual Agent intake and SMS or Teams notifications.
- Predictive intelligence or an ML text classifier once enough ticket data exists.
- Dashboards and reports on ticket volume by category.

11. Appendix

Item	Details
Source Code	Not applicable - no-code solution built with Flow Designer. The configuration is in the exported update set 'Project Update Set' (sys_remote_update_set_a7b40de6fc590ceaad36c.xml) .
Dataset Link	Not applicable - tickets are created in the ServiceNow table u_incident_workflow.
GitHub & Project Demo Link	[Add your GitHub repository / demo video link here]